TUTORIAL

A simple SN2 reaction with a QC method

> Updated for EasyHybrid 3.2.0 (2026-10-26). The figures and numbers come from a run of this tutorial with the 3.2.0 development version.

In a bimolecular nucleophilic substitution (SN2) the nucleophile attacks the carbon from the side opposite to the leaving group; the bond to the nucleophile forms while the bond to the leaving group breaks, through a single transition state with the three groups of the carbon in a plane:

$Br^- + CH_3{-}Cl \rightarrow [Br\cdots CH_3\cdots Cl]^- \rightarrow Br{-}CH_3 + Cl^-$

In this tutorial you will:

1. load a structure without force field and describe all its atoms with a semiempirical QC method (PM6);
2. optimize the reactant complex;
3. scan a **multiple distance** reaction coordinate, the usual coordinate of a group transfer;
4. find the transition state and the product complex in the energy profile.

The run takes less than a minute. The structure is `examples/tutorials/02_sn2/SN2_tutorial.xyz` in the EasyHybrid folder (6 atoms: CH_3Cl and a bromide ion 3.1 Å from the carbon).

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1. Load the structure

File → New:

Field	Value
Name / Tag	`SN2` / `PM6`
System Type	**other (*.pdb, .xyz, *.mol2)**: the structure is loaded without force field
Files	`examples/tutorials/02_sn2/SN2_tutorial.xyz`

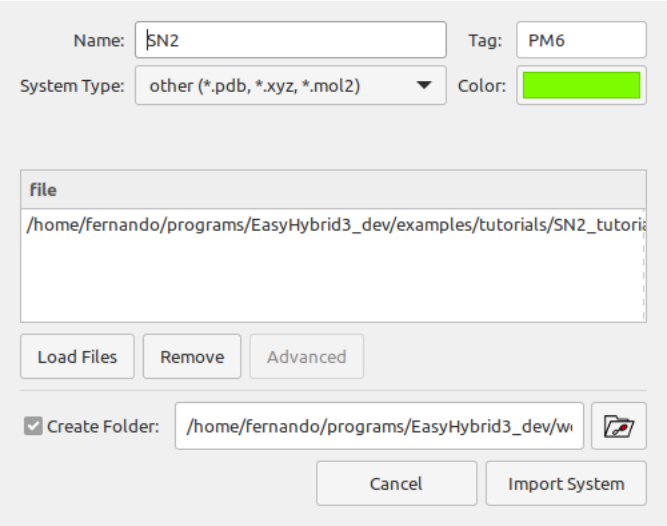


Figure 1. A system without force field.

Click **Import System**. A system without force field cannot be simulated until it has an energy model: here, a QC model for all its atoms.

2. The QC model

Press **Quantum Chemistry** in the toolbar (**System → Setup QC Region**). With no atoms selected, the QC model covers the whole system: **Number of Atoms** shows `6 (all)`.

Field	Value
QC Engine	pDynamo
Method	**pm6**
Charge	**−1** (the bromide)

Field	Value
Multiplicity	1
Restricted	on (closed shell)

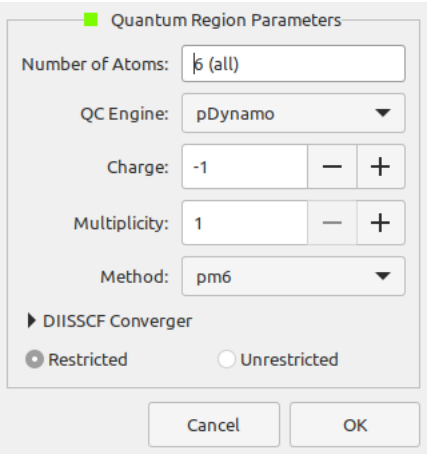


Figure 2. PM6 for the six atoms, total charge −1.

Click **OK**. EasyHybrid confirms that the QC model was defined; the atoms are now drawn as balls and sticks, the representation of QC atoms, and the status bar shows `hamiltonian: pm6`, `QC charge: -1`.

3. Optimize the reactant complex

Simulate → Geometry Optimization: Conjugate Gradient, 1000 iterations, RMS gradient 0.01, and **Run**. The optimized ion–molecule complex has C–Cl = 1.82 Å and C···Br = 3.09 Å, with the bromide on the axis of the C–Cl bond, opposite to the chlorine.

4. The reaction coordinate

The reaction breaks C–Cl and forms C–Br at the same time. One distance alone is a poor coordinate (pulling the bromide in pushes the whole molecule); the difference of the two distances follows both bonds:

RC = d(Cl–C) − d(C–Br)

It is negative at the reactant (1.82 − 3.09), near zero at the transition state and positive at the product.

1. Hide the first object (`0_coords`) and switch to **Picking** mode.
2. Pick, in this order, **Cl** (#1), **C** (#2) and **Br** (#3). The central atom, the one transferred between the other two, must be #2.

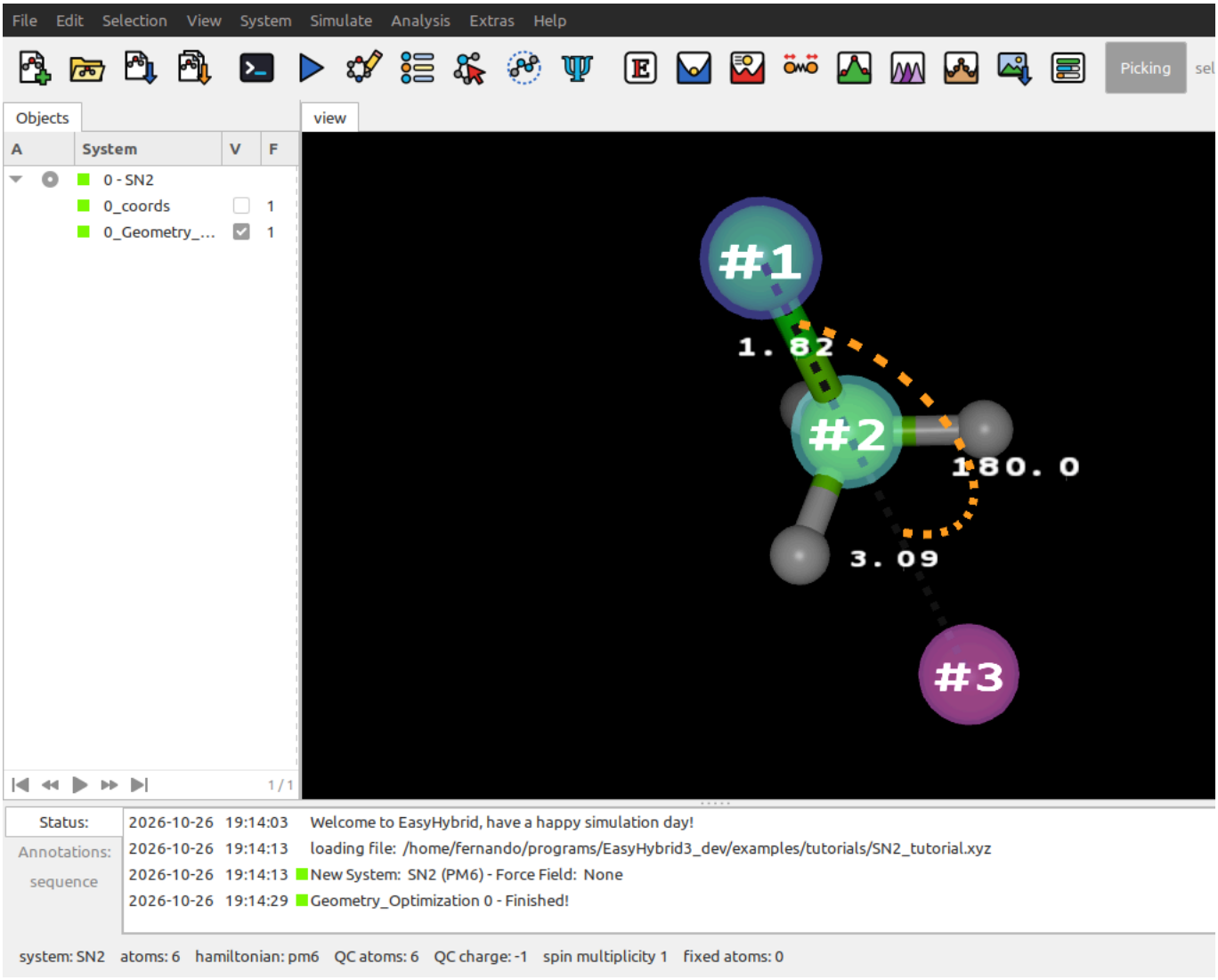


Figure 3. Cl (#1), C (#2) and Br (#3); the distances and the Cl–C–Br angle are shown.

5. Scan

Simulate → Reaction Coordinate Scans. In **Reaction Coordinate 1** choose the **Coordinate Type** *multiple distance* and click **Import from Picking Selection**:

Field	Value
Atoms 1, 2, 3	Cl2, C0, Br5 (indices 2, 0, 5)
Apply Mass Weighted Restraints	on (set automatically: atoms 1 and 3 are different elements)
Step Size	0.1 Å
Number of Steps	14
Force Constant	4000 kJ mol^{-1} $Å^{-2}$
Coordinates	the optimized structure
Method / Maximum iterations / RMS Grad. Tolerance	Conjugate Gradient / 1000 / 0.1
Trajectory Name	`SN2_scan`

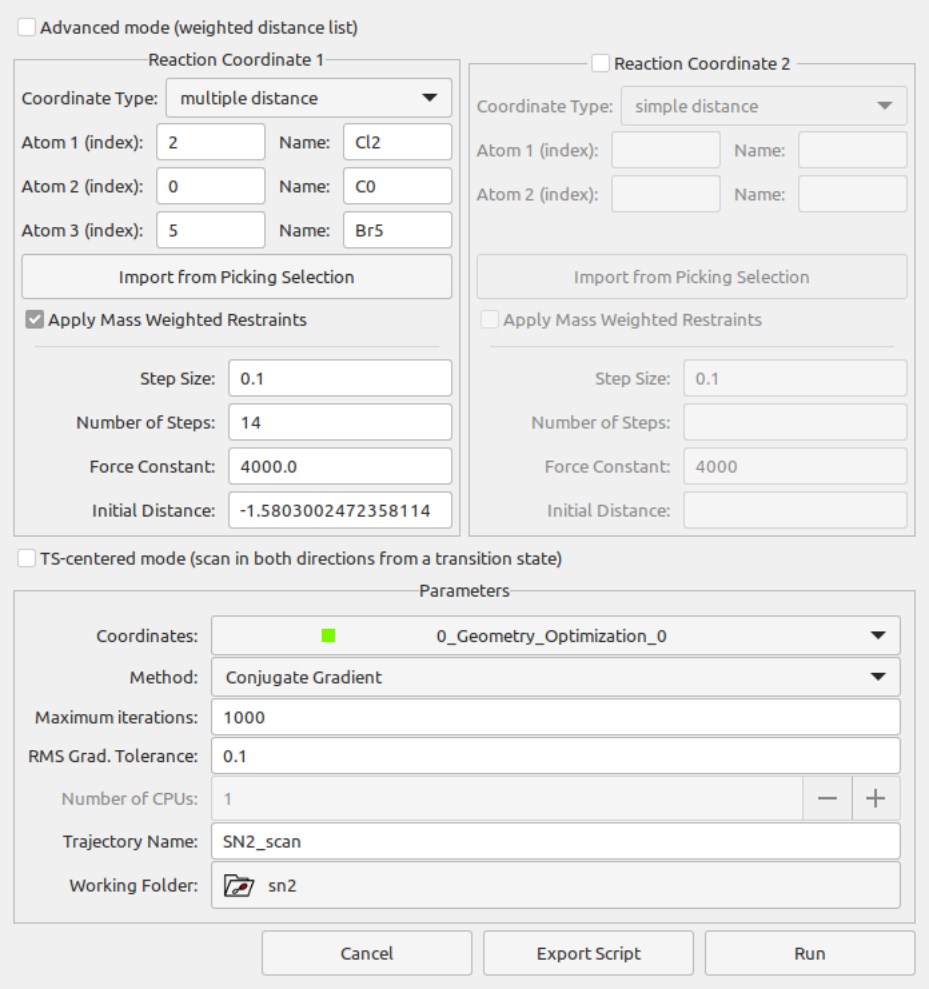


Figure 4. The multiple-distance coordinate Cl–C–Br.

Mass-weighted restraints. When the two end atoms have different masses, the two distances are weighted by the masses of Cl and Br (0.31 and 0.69 here), so that the restraint moves the light and the heavy end in proportion to their masses. The **Initial Distance** is then the weighted value (−1.58 Å here), not the plain difference.

Click **Run** and, when it finishes, **Import…** in the *Finished!* dialog and **Import**.

6. The energy profile

Analysis → PES Analysis, with the scan object:

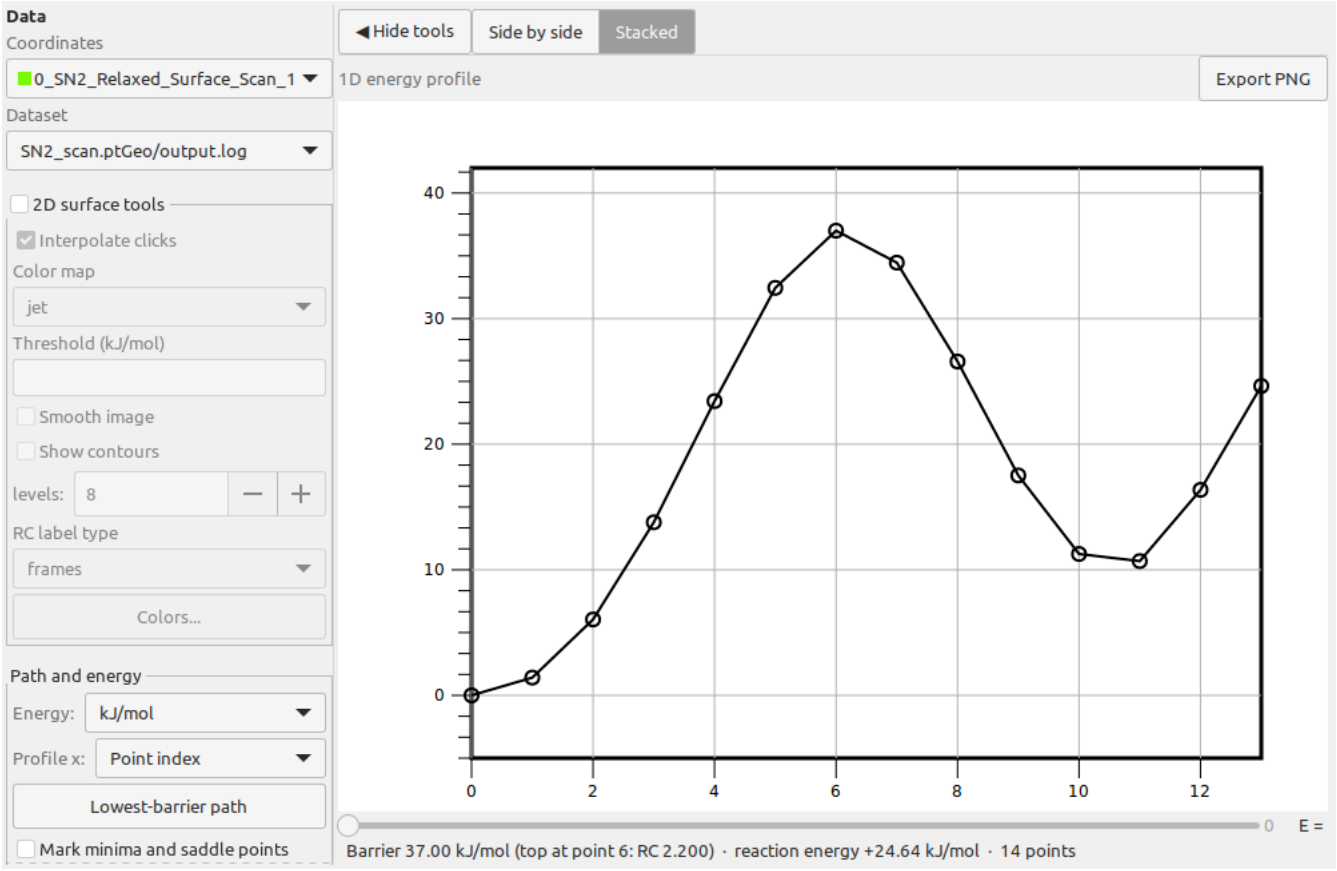


Figure 5. Energy along the scan (relative to the reactant complex).

Point	d(Cl–C) (Å)	d(C–Br) (Å)	E (kJ/mol)	
0	1.82	3.09	0.0	reactant complex $Br^{-}\cdots CH_3Cl$
6	2.20	2.39	**37.0**	transition state region
11	2.88	1.98	10.7	product complex $BrCH_3\cdots Cl^{-}$
13	3.41	1.93	24.6	Cl^{-} leaving the complex

The highest point (point 6) has the carbon between the two halogens and its three hydrogens almost in a plane (Figure 6): the Walden inversion. The product complex is 10.7 kJ/mol above the reactant complex: in the gas phase chloride is the stronger nucleophile and $Cl^{-} + CH_3Br \rightarrow CH_3Cl + Br^{-}$ is the downhill direction, so this reaction, written the other way, is uphill. Beyond point 11 the scan only pulls the chloride away from the product complex.

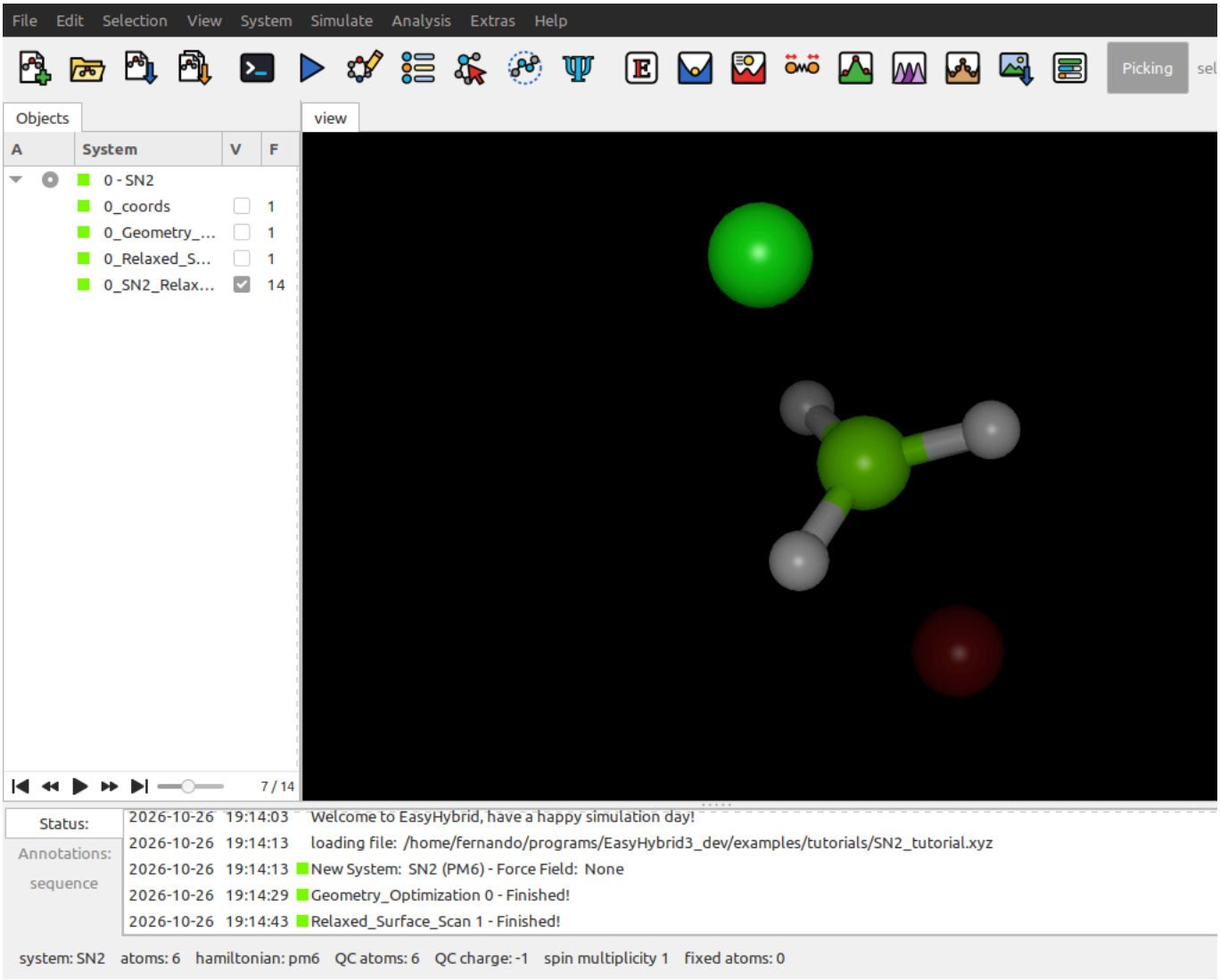


Figure 6. Point 6, the highest point of the profile.

Notes

- **The highest point of a scan is not the transition state**, only close to it. Refine it with **Simulate → Search Transition State (Baker)**, starting from point 6, and check that it has a single imaginary frequency (**Simulate → Normal Modes**).
- **Step size.** 0.1 Å is enough to locate the barrier; a finer scan (0.05 Å) around the top gives a better estimate.
- **Gas phase.** Ion–molecule reactions in the gas phase have deep complexes on both sides and low barriers; in water the barriers are much higher (the solvent stabilizes the separated ions).
- Reference: M. L. Chabinyc, S. L. Craig, C. K. Regan, J. I. Brauman, *Science* **1998**, 279, 1882; S. Schmatz, D. C. Clary, *J. Chem. Phys.* **1999**, 110, 9483 (dynamics of Cl^- + CH_3Br).

EasyHybrid 3.2.0 · built 2026-10-29 from `dev` @ `3539da4` · https://github.com/ferbachega/EasyHybrid3